

Secondary Data Analysis (Project 1) – Overview and Plan

Basic Information of the dataset:

Name of the Dataset: Sleep Deprivation & Cognitive Performance

Publication Year: The public dataset is published on Kaggle in 2024.

URL : (<https://www.kaggle.com/datasets/sacramentotechnology/sleep-deprivation-and-cognitive-performance>) (DOI : <https://doi.org/10.34740/kaggle/dsv/10569288>)

License: CC 1.0 Universal (Public Domain). The dataset is available for use, modification, and redistribution. The original Kaggle source will nevertheless be acknowledged for research transparency.

Software Used for Analysis: RStudio (2026.01.0 +392)

Study Variables: Sleep-related variables (e.g. sleep hours, sleep quality). Emotion regulation scores (higher scores = better emotion regulation). Scores on cognitive performances. Age and gender. BMI, caffeine intake, physical activity level and stress level Each observation is independent. (N = 60)

Research Question and Plan

The project aims to evaluate two questions relating to psychology and cognitive neuroscience.

Aim 1: The first aim investigates the relationships between sleep duration, sleep quality, and emotional-regulation performance. It also examines whether sleep quality moderates the relationship between sleep duration and emotional regulation.

Aim 2: The second aim investigates the relationships between psychological or sleep-related measurements and three cognitive-task outcomes. An exploratory PCA will first examine whether these outcomes can be compressed into a coherent overall cognitive-performance index. If no sufficiently dominant and interpretable principal component is identified, the three task outcomes will be analyzed separately.

Additionally, the two relationships have been theoretically discussed in plenty of previous studies; therefore, the data analysis project also aims to further examine the previous findings. Overall, the analyses are exploratory, and theory guided.

Predictors, Covariate and Outcome

Aim 1: Outcome: Emotion-regulation score

Predictors: Sleep duration and sleep quality

Aim 2: Outcomes: Stroop RT, N-back accuracy, and PVT RT

Predictors/covariates: Emotion regulation, stress group, sleep duration, and sleep quality

Potential Techniques and Models:

Aim1: (1) Simple Linear regression (3) Multiple Linear Regression

Aim2: (1) Principal Component Analysis (PCA) (2) ANCOVA (3) Simple linear regression and multiple linear regression

Data Preparation

Prior to the formal analysis, the dataset will be initially screened and checked for missing values. And for facilitation of interpretation, the mean-centering will be applied for interaction effect analysis in study 1 because it can give practically reasonable interpretation to lower-order coefficients.

Model Assumptions Diagnostic

Validity of inference from theoretically expected sampling distributions relies heavily on meeting the assumptions. Hence, checks of regression assumptions and model diagnostics will be conducted after each model is fitted.

- (1) Approximate normality of residuals
- (2) Homoscedasticity (homogeneity of variance)
- (3) Linearity of the modeled relationship
- (4) Multicollinearity in models containing multiple predictors
- (5) Influential outliers will be checked as necessary for the models.

Since the authors have declared that all measurements come from separate individuals, it is expected that all observations are separate, and they will therefore be treated as independent. Therefore, a formal check for autocorrelation is not considered necessary.